

When Does Success Breed Success for Women?

Evidence from Elite Table Tennis

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Abstract

A consensus in the economics of sport holds that winning breeds further winning for men but not for women, with post-victory testosterone the favoured explanation. I show that this consensus is sport-specific. Using 12,227 elite table tennis matches played between 2014 and 2026, I treat deuce sets (tied at 10–10) as a natural experiment in which the eventual winner is as good as randomly assigned. Winning a deuce set raises the probability of winning the next set by 8.8 percentage points, an 18% gain over the 49.5% baseline. The effect is more than twice as large for women (a 27% gain over their female baseline) as for men (an 11% gain over their male baseline), and the female advantage widens further with each additional point played beyond 10–10. Point-level streak evidence shows that the female advantage is absent at the shortest streaks and builds up as competitive pressure accumulates, ruling out mechanical artefacts. The reversal fits a two-pathway view of momentum: in technique-intensive sports the physiological channel that favours men is muted, leaving a cognitive-affective channel that favours women.

Keywords: psychological momentum, gender, close contest, natural experiment, table tennis

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1 Introduction

A robust finding in the economics of sport is that winning breeds further winning, but only for men. Studies of judo medal fights (Cohen-Zada et al., 2017) and professional tennis (Page and Coates, 2017; Gauriot and Page, 2019) find a precise male momentum effect alongside a female point estimate indistinguishable from zero. The standard explanation appeals to testosterone, which surges after victory in men but not in women (Cohen-Zada et al., 2017). Related work traces the effect to psychological rather than strategic channels (Meier et al., 2020; Descamps et al., 2022; Kubitz et al., 2025).

Yet every one of these studies uses a sport in which physical power and aggression play a substantial role. A separate literature shows that gender differences in competitive behaviour are highly task-specific: they shrink or reverse in stereotypically female domains (Wieland and Sarin, 2012), disappear among children in certain physical tasks (Dreber et al., 2011), and reverse entirely in matrilineal societies (Gneezy et al., 2009). If the sign of the gender gap in *willingness to compete* is task-dependent, why should the sign of the gender gap in *responses to competitive feedback* not be? A complementary literature documents the opposite pattern once selection is held fixed: women are *more* sensitive to competitive pressure than men. Female students in China underperform on the high-stakes college entrance examination by about 0.15 standard deviations relative to their own mock results (Cai et al., 2019). The gender gap in a Spanish mathematics contest widens by roughly fifty per cent between a qualifying round and a more pressured final (Iriberry and Rey-Biel, 2019). And in tennis, women who lose the first set (De Paola and Scoppa, 2017) or play a closely contested serve game (Lackner and Weichselbaumer, 2023) subsequently underperform comparable men. Together these findings yield a sharp conjecture: in a setting where the physiological channel is muted, women's greater responsiveness to competitive feedback should *reverse* the conventional gender pattern of momentum.

This paper tests that conjecture. I study psychological momentum in elite table tennis, a technique-intensive sport in which outcomes depend on reaction speed, fine motor control, and mental composure rather than on strength or stamina. I exploit deuce sets, tied at 10–10, as a natural experiment.

When a set reaches 10–10, both players have just traded twenty points under identical conditions, so the two-point margin that follows is as good as randomly assigned. This close-contest identification strategy follows [Cohen-Zada et al. \(2017\)](#), [Page and Coates \(2017\)](#), and [Gauriot and Page \(2019\)](#). The dataset covers 12,227 professional singles matches sanctioned by the International Table Tennis Federation (ITTF) and World Table Tennis (WTT) between 2014 and 2026, featuring players from 115 countries and yielding 7,501 deuce sets.

I document four main results. First, winning a deuce set raises the probability of winning the next set by 8.8 percentage points ($p < 0.001$), an 18% gain over the 49.5% baseline. Second, the effect is more than twice as large for women (12.9 pp, a 27% gain on the female baseline) as for men (5.7 pp, an 11% gain); the female–male difference is 7.1 pp ($p = 0.007$) and survives restriction to first-set deuces, where the quasi-random comparison is cleanest. Third, the gender gap widens with deuce duration: each additional point played beyond 10–10 widens the female advantage by 2.1 pp ($p = 0.018$), so that long, high-pressure deuces erode men’s momentum while amplifying women’s. Fourth, point-level streak evidence shows that the female advantage is absent at the shortest within-set streaks, ruling out mechanical artefacts of serve rotation or scoring, and builds up as competitive pressure accumulates.

Two pathways link winning to subsequent performance, and which pathway dominates depends on the task. A *physiological* pathway operates through post-victory testosterone and favours men in tasks where physical aggression matters ([Cohen-Zada et al., 2017](#)). A *cognitive-affective* pathway operates through confidence, belief updating, and affective responses to feedback ([Kubitz et al., 2025](#); [Descamps et al., 2022](#); [Gill and Prowse, 2012](#)). In technique-intensive tasks the physiological channel is muted, leaving the cognitive-affective channel to dominate. Because women’s competitive responses are more context-dependent and more feedback-sensitive than men’s ([Croson and Gneezy, 2009](#); [Niederle and Vesterlund, 2011](#)), the cognitive-affective channel favours women. The reversal in table tennis, paired with converging evidence on negative momentum ([De Paola and Scoppa, 2017](#); [Lackner and Weichselbaumer, 2023](#)) and on choking in other technique-intensive sports ([Bühren et al., 2024](#)), fits this two-pathway logic.

This paper makes three contributions. First, it provides the first evidence of *positive* momentum from a sport in which physical power is not the binding constraint. The male-only finding of [Cohen-Zada et al. \(2017\)](#), [Page and Coates \(2017\)](#), and [Gauriot and Page \(2019\)](#) does not generalise beyond physical sports. Second, it adds a real-stakes field setting to the literature on gender and competition ([Gneezy et al., 2003](#); [Niederle and Vesterlund, 2007, 2011](#); [Croson and Gneezy, 2009](#); [Wieland and Sarin, 2012](#)), showing that women’s heightened sensitivity to competitive feedback operates not only after losses ([Cai et al., 2019](#); [De Paola and Scoppa, 2017](#); [Lackner and Weichselbaumer, 2023](#)) but also after wins. Third, pairing set-level estimates with point-level streak evidence and a deuce-duration dose–response narrows the space of mechanisms to those that accumulate under competitive pressure.

Section 2 describes the setting and data, Section 3 presents the empirical strategy and results, Section 4 discusses mechanisms, and Section 5 concludes.

2 Setting, Natural Experiment, and Data

2.1 Elite Table Tennis and the Deuce Rule

Professional table tennis is governed by the ITTF and its commercial arm, WTT. Matches are played in best-of-five or best-of-seven set format, with each set played to 11 points. A player must win by two clear points; if the score reaches 10–10 (“deuce”), play continues until one player leads by two.

The deuce rule creates a natural experiment for studying momentum. Selection into reaching 10–10 is of course not random: mismatched pairs arrive at deuce through very different paths than evenly matched ones. But *conditional* on having reached 10–10, both players have just traded twenty points under identical conditions of venue, equipment, and opponent. The two or more additional rallies that follow then decide the set by a margin of exactly two points. Conditional on reaching deuce, the identity of the winner is as good as randomly assigned: small variations in execution, luck, and nerve decide a contest between players who have just demonstrated parity. This close-contest logic follows [Cohen-Zada et al. \(2017\)](#), [Page and Coates \(2017\)](#), and [Gauriot](#)

and Page (2019). Appendix 5 verifies the underlying balance assumption.

Table tennis suits this question for two reasons. First, the sport is technique-intensive: rallies last a few seconds, the ball weighs 2.7 grams, and points are decided by reaction speed, fine motor control, and mental composure rather than by strength or stamina. The physiological pathway that prior work invokes to explain male-only momentum in judo and tennis should be weaker here. Second, men's and women's competitions run under identical rules, formats, and scheduling on the same tours, so any male–female gap is not contaminated by differences in stakes or institutional environment.

2.2 Data

I construct a set-level panel from 12,227 professional singles matches played between January 2014 and March 2026,¹ yielding 49,992 set-level observations. Match data come from Sofascore,² which provides official point-by-point results for ITTF- and WTT-sanctioned events, including the World Championships, WTT Champions, WTT Smash, WTT Star Contender, WTT Contender, and WTT Feeder events. A random audit re-scrapes 99 matches and achieves 100% agreement with the working dataset. For each match I designate the player listed first on the platform as the reference player, and all regressions adopt that player's perspective.

The estimating sample narrows in two steps. Of the 49,992 sets, 7,501 (15.0%) reach deuce. Dropping final-set deuces, which have no subsequent set, leaves 5,773 deuce sets in 4,713 unique matches: 2,467 deuce sets in women's matches and 3,306 in men's matches. Table 1 presents summary statistics. Deuce sets average 24.1 total points against 18.4 for all sets. The reference-player win rate in deuce sets is 49.5%, indistinguishable from 50% (binomial test $p = 0.43$). Among deuce sets with a subsequent set, the set winner wins the next set 53.0% of the time, compared with 59.0% for all sets. Deuce outcomes thus exhibit the smallest carry-over of any set type. This is exactly what quasi-randomness should produce: it washes out the ability-based selection that

¹The sample ends on 31 March 2026; coverage of 2026 therefore comprises the first quarter of the season only.

²<https://www.sofascore.com>

normally determines who wins a set.

Table 1: Summary Statistics

Panel A: Sample overview

Set-level observations	49,992
Unique singles matches	12,227
Unique players	1,696
Years covered	2014–2026
Female sets	22,995
Male sets	26,983

Panel B: Set characteristics

	All sets	Deuce sets
Observations	49,992	7,501
Mean total points	18.4	24.1
SD total points	3.3	2.9
Mean point margin	4.2	—
Win rate	0.516	0.495
Winner wins next set	0.590	0.530

Panel C: Deuce duration distribution

Short deuce (12–10)	3,684
Medium deuce (13–11)	1,903
Long deuce (14–12+)	1,911
Total deuce sets	7,501

Notes. Win rates and *Winner wins next set* are from the reference player’s perspective (Sofascore listing order).

3 Empirical Strategy and Results

3.1 Empirical Strategy

The identification strategy exploits the quasi-randomness of deuce outcomes. I estimate the effect of winning a deuce set on next-set performance from the linear probability model

$$Y_{ij} = \alpha + \beta \text{Won}_{ij} + \gamma \text{SetsAhead}_{ij} + \delta' X_{ij} + \varepsilon_{ij}, \quad (1)$$

where Y_{ij} indicates whether the reference player wins the next set, Won_{ij} indicates whether she won the deuce set, SetsAhead_{ij} is the set-score advantage entering the deuce, and X_{ij} collects indicators for a female match and for Chinese involvement. The subscript i indexes the deuce set and j the match. Standard errors cluster at the match level. To test gender heterogeneity I interact Won_{ij} with a female-match dummy. To test whether the gap varies with deuce intensity, I add the triple interaction $\text{Won}_{ij} \times \text{Female}_{ij} \times \text{Duration}_{ij}$, where Duration_{ij} is the number of points played beyond 10–10; all lower-order interactions and SetsAhead_{ij} are retained as controls.

Appendix 5 verifies the identifying assumption. The reference-player win rate in deuce sets is indistinguishable from 50% across rounds, years, and first-set subsamples. The balance regression shows that *SetsAhead* is itself predictive of deuce-set outcomes, because players already leading the match are more likely to close the deuce set, so the main specification conditions on it. Once *SetsAhead* is held fixed, neither *Female* nor *China-involved* predicts the deuce outcome, and in the cleanest first-set subsample (where *SetsAhead* is mechanically zero) *Female* has a point estimate of 0.004 (SE = 0.028).

3.2 Main Effect and Gender Heterogeneity

Table 2 reports the paper’s central results. Panel A gives the pooled main effect of winning a deuce set. Column (1) shows a raw effect of 6.0 percentage points. Column (2) adds controls for sets ahead, gender, and Chinese involvement, raising the estimate to 8.8 pp ($p < 0.001$), an 18% gain

over the 49.5% baseline. Set-number fixed effects leave the estimate virtually unchanged at 8.7 pp (Column 3), and the effect is 8.9 pp in the first-set-only subsample where *SetsAhead* is mechanically zero and the quasi-random comparison is cleanest (Column 4).

Panel B disaggregates the effect by gender. Column (1) introduces a $\text{Won} \times \text{Female}$ interaction: the base effect for men is 5.7 pp ($p = 0.001$) and the additional effect for women is 7.1 pp ($p = 0.007$). Columns (2) and (3) estimate the model separately by gender and report 12.9 pp for women ($p < 0.001$) versus 5.7 pp for men ($p = 0.001$). Relative to each gender's baseline, the female effect is a 27% gain on the 0.484 female baseline, whereas the male effect is an 11% gain on the 0.502 male baseline. Women's momentum is not merely larger in absolute terms: it is more than twice as large in relative terms. This gender gap carries through to match-level outcomes: women gain an additional 4.8 pp in match-win probability relative to men (Appendix Table A4).

Column (4) of Panel B adds the triple interaction $\text{Won} \times \text{Female} \times \text{Duration}$. The triple coefficient is +0.021 ($p = 0.018$): each additional point beyond 10–10 widens the female–male momentum gap by 2.1 pp. For men, the $\text{Won} \times \text{Duration}$ coefficient is -0.011 ($p = 0.089$); summing the two gives a female-specific slope of +0.010. Longer deuces therefore flip the male slope into negative territory while keeping the female slope positive. The triple coefficient captures this divergence and is the testable object. Aggregating to the match level (Appendix Table A4, Column 5) attenuates the triple interaction to insignificance, consistent with the lower signal-to-noise ratio of binary match outcomes; the gender main effect, however, survives (Column 4).

Table 2: Momentum and Gender Heterogeneity

Panel A. Pooled main effect				
	(1) Raw	(2) + Controls	(3) + Set FE	(4) Set 1 only
<i>Dep. var.: Won next set</i>				
Won deuce set	0.060*** (0.013)	0.088*** (0.013)	0.087*** (0.013)	0.089*** (0.024)
Sets ahead control	No	Yes	Yes	—
Demographics	No	Yes	Yes	Yes
Set number FE	No	No	Yes	—
DV mean	0.495	0.495	0.495	0.493
Observations	5,773	5,773	5,773	1,784
Panel B. Gender heterogeneity				
	(1) Interaction	(2) Female	(3) Male	(4) Triple
<i>Dep. var.: Won next set</i>				
Won deuce set	0.057*** (0.017)	0.129*** (0.020)	0.057*** (0.017)	0.100*** (0.030)
Won × Female	0.071*** (0.026)			-0.015 (0.045)
Won × Duration				-0.011* (0.006)
Won × Female × Duration				0.021** (0.009)
Lower-order terms	Yes	—	—	Yes
Sets ahead control	Yes	Yes	Yes	Yes
Demographics	—	Yes	Yes	—
DV mean	0.495	0.484	0.502	0.495
Observations	5,773	2,467	3,306	5,773
R-squared	0.024	0.031	0.018	0.025

Notes. Sample: deuce sets with a subsequent set. Dependent variable: indicator for winning the next set. Standard errors in parentheses, clustered by match. * p<0.10, ** p<0.05, *** p<0.01.

Appendix 5 confirms that the main effect remains highly significant across alternative specifications (first sets only, heteroskedasticity-robust standard errors without clustering, best-of-five matches only, and samples excluding matches involving Chinese players), with point estimates ranging from 7.4 to 8.9 pp. Appendix 5 reports estimates with player fixed effects that absorb time-invariant ability.

3.3 A Reversal of the Gender Consensus

The central finding is not only that momentum exists in table tennis: its gender ordering flips the prior literature's sign. The two most directly comparable studies in economics both find momentum concentrated among men. [Cohen-Zada et al. \(2017\)](#), who estimate momentum at the match level in judo using bronze-medal fights, report roughly +21 pp for men and a point estimate close to zero (and insignificant) for women. [Gauriot and Page \(2019\)](#), who estimate point-level momentum in tennis using a fuzzy regression discontinuity on balls landing near the sideline, report +7.2 pp for men ($p = 0.012$) and +1.2 pp for women ($p = 0.74$). The present set-level estimates invert this gender ordering: +5.7 pp for men versus +12.9 pp for women. Moving from a physical sport (judo) and a semi-physical sport (tennis) to a technique-intensive sport (table tennis), the male advantage flips into a female advantage. Absolute magnitudes do not compare cleanly across studies because the units differ (match, point, and set), but within-study male–female gaps are directly comparable and they point in opposite directions.

3.4 Dose–Response by Gender

Longer deuces erode men's momentum and amplify women's. Figure 1 visualises the dose–response pattern implied by the triple interaction. The set winner's next-set win probability for men declines from 53.3% at short deuces (12–10) to 48.8% at long deuces (14–12 or beyond), crossing the 50% no-advantage threshold. For women it *ris*es from 54.4% to 57.3% over the same range. Longer, more pressured deuces therefore push men *away* from positive momentum and women *towards* it. This is the central empirical pattern of the task-type reversal.

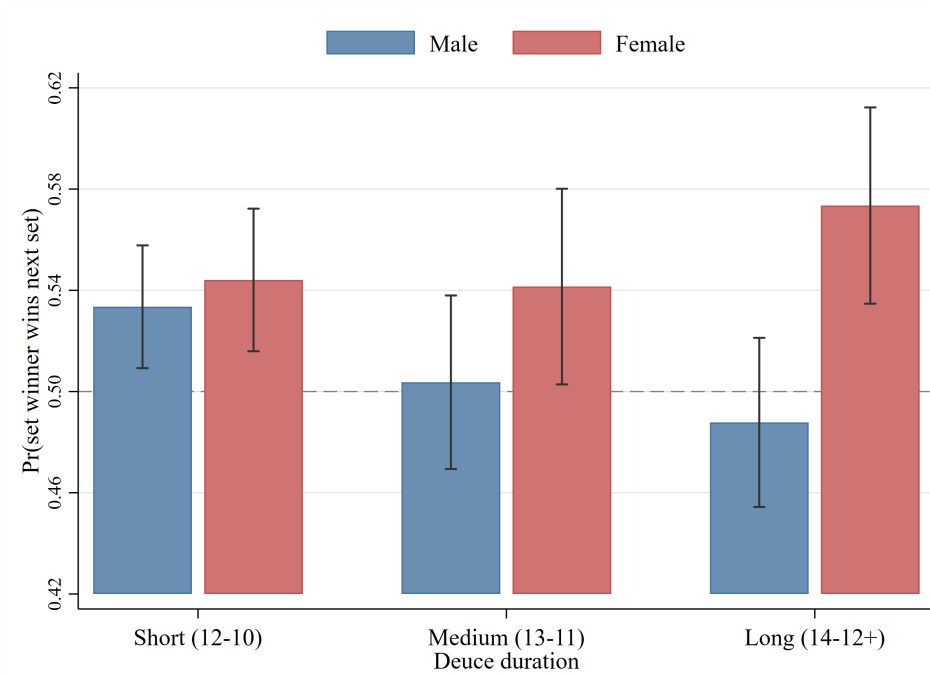


Figure 1: Momentum by Gender and Deuce Duration

Notes. Each bar reports the set winner's probability of winning the next set, by deuce duration and gender. Error bars show 95% confidence intervals. The dashed line marks 0.50. Sample: deuce sets with a subsequent set.

3.5 Point-Level Evidence on the Proposed Channel

The set-level results establish the reversal but are silent on the within-set dynamics that generate it. I therefore pair them with point-by-point evidence. The goal is not to pin down the precise psychological mechanism but to ask whether the within-set patterns line up with a cognitive-affective channel that accumulates under pressure, or with a mechanical or purely physiological one.

The female advantage is not present from the first favourable point; it builds up. Figure 2 shows the unconditional streak-continuation curve. Let k denote the length of the current within-set winning streak, and let $\Pr(\text{win next point} \mid k)$ denote the probability that the streak extends by one more point. For men the curve is flat in k : winning one, two, or three consecutive points barely shifts the probability of winning the next. For women it rises in k , so that a player on a three-point streak is notably more likely to extend it than a player at $k = 1$. The male-female divergence is small at $k = 1$, where only a single favourable shock has accumulated, and grows with k . This is the point-level analogue of the set-level dose-response: the female advantage builds up as competitive

pressure accumulates.

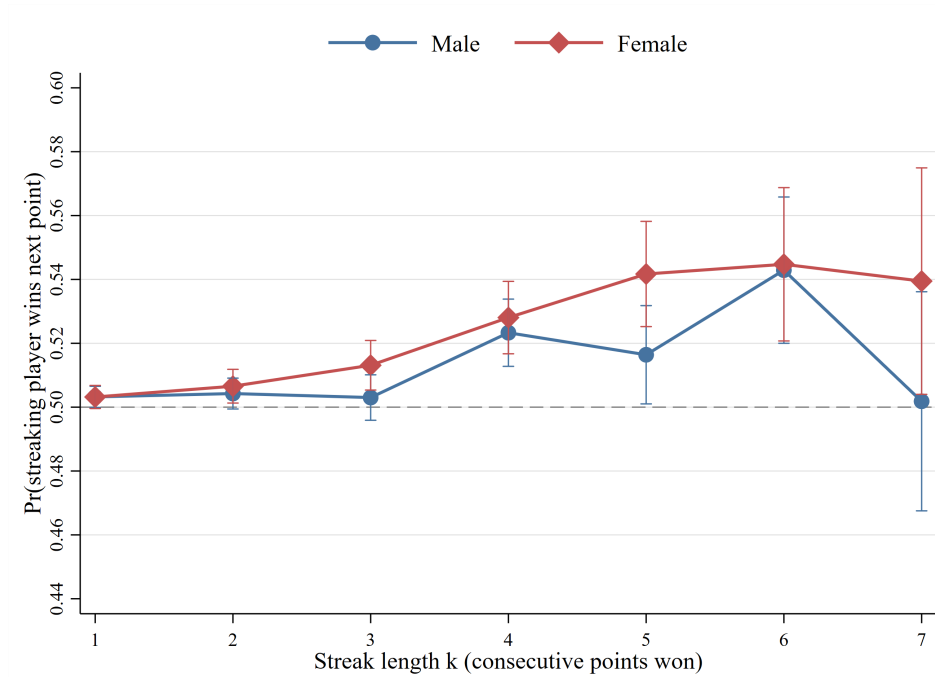


Figure 2: Point-Level Streak Continuation by Gender

Notes. Probability that the current within-set winning streak is extended by one point, as a function of streak length k , separately for men and women. Error bars show 95% confidence intervals. The dashed line marks 0.50. Sample: all points in deuce sets.

Two supplementary exercises support this reading. First, Appendix Table A5 re-estimates a point-level linear probability model restricted to $k = 1$, the shortest possible streak. A mechanical artefact (serve rotation, scoring rules, or automatic mean reversion) should appear at $k = 1$ just as strongly as at longer streaks. It does not: at $k = 1$ the gender gap is small and indistinguishable from zero, ruling out purely mechanical explanations. Second, Appendix Table A6 partitions deuce sets by the length of their *closing streak*, defined as the number of consecutive final points the eventual set winner took immediately before closing the set. Within each bin, the table reports the female–male gap in next-set win probability. A female advantage appears in the “battle” bin ($k = 1$, +9.5 pp, $p < 0.05$), is absent in the middle bins ($k = 2, 3$), and reappears in the “dominant” bin ($k \geq 4$, +11.3 pp, $p < 0.10$, $N = 241$). The pattern is not cleanly monotone, but it shows that the female advantage survives conditioning on the within-set closing structure rather than being an artefact of the unit of analysis.

4 Mechanisms and Discussion

The set-level reversal and its dose–response pattern point to a channel that accumulates with competitive pressure rather than to one tied to physical aggression. A two-pathway framework rationalises this pattern.

Two pathways. A *physiological* pathway operates through post-victory testosterone, which surges in men but not women and sharpens physical aggression (Cohen-Zada et al., 2017). A *cognitive-affective* pathway operates through confidence, belief updating about relative ability, the subjective value of the prize, and affective responses to feedback (Kubitz et al., 2025; Descamps et al., 2022; Gill and Prowse, 2012); under high stakes it can also impair execution through choking on automatised skills (Baumeister, 1984; Ariely et al., 2009). Both pathways are likely present in both sport types: the task-type prediction is about which dominates, not which is uniquely operative. Meier et al. (2020) confirm that even in men’s professional tennis, where the physiological story is strongest, the cognitive-affective channel is independently live.

In physical sports such as judo and tennis, the physiological pathway amplifies a male advantage. In table tennis it is muted, leaving the cognitive-affective pathway to dominate. Because women’s competitive responses are more context-dependent and feedback-sensitive than men’s (Croson and Gneezy, 2009; Niederle and Vesterlund, 2011), this pathway favours women. Bühren et al. (2024) document a parallel regularity in alpine skiing: in the decisive second slalom run, a compatriot course-setter raises men’s non-completion rate and lowers women’s.

Three candidate mechanisms. Three mechanisms within the cognitive-affective pathway can generate the observed asymmetry; the data do not let me distinguish them.

First, *feedback-sensitive effort*. Women’s effort may respond more elastically than men’s to competitive feedback in both directions. The negative-momentum literature shows women underperform more than men after losing a contested set (De Paola and Scoppa, 2017; Lackner and Weichselbaumer, 2023) or under high-stakes pressure (Cai et al., 2019; Iriberry and Rey-Biel, 2019).

Symmetry across the sign of the shock then predicts a stronger female response to a contested *win*, matching the positive asymmetry documented here. Consistent with this, the female advantage is strongest in the longest deuces, precisely where competitive feedback is most intense.

Second, *arousal and self-focus*. Skill performance is non-linear in arousal: moderate arousal helps, excess triggers self-focus that disrupts automatised movements (Baumeister, 1984; Ariely et al., 2009). If men enter competition with higher baseline confidence (Niederle and Vesterlund, 2007; Croson and Gneezy, 2009), a further boost from winning may push them past their effective threshold while moving women towards theirs. This fits the evidence that men choke more often in technique-intensive settings (Bühren et al., 2024) and that interim leaders in dynamic tournaments take more risk but succeed less (Genakos and Pagliero, 2012).

Third, *belief updating*. Winning can reveal favourable information about relative ability that shifts effort and strategy (Descamps et al., 2022). If women hold more conservative priors (Niederle and Vesterlund, 2007), a contested win is a larger positive signal relative to expectations. But Wozniak et al. (2014) find that explicit performance feedback closes the gender gap in tournament entry, suggesting belief frictions alone cannot carry the full story.

Dose–response as an internal check. The triple interaction is hard to reconcile with a purely strategic channel: longer deuces raise psychological intensity but do not alter strategic incentives at a given deuce score. That they amplify women’s momentum while eroding men’s instead fits a cognitive-affective channel that accumulates with the duration of competitive uncertainty. It also matches the arousal-threshold prediction directly.

Converging evidence. Women respond more strongly than men to competitive feedback in *both* directions, the asymmetry the task-type interpretation requires (Cai et al., 2019; Iriberry and Rey-Biel, 2019; De Paola and Scoppa, 2017; Lackner and Weichselbaumer, 2023). The newest piece of evidence is non-athletic: Spilker et al. (2025) show that female football referees, especially younger ones, award significantly less stoppage time than male referees specifically when the home team is narrowly leading (the situation in which crowd pressure to end the match is highest), while no gender

gap arises in tied or away-leading games. The prior literature's conclusion that positive momentum is absent for women in tennis and judo may therefore reflect the dominance of the physiological pathway in those settings, which masks a cognitive-affective pathway that favours women.

Limitations. Four limitations deserve mention. First, identification relies on deuce outcomes being quasi-random conditional on reaching 10–10; the balance tests in Appendix 5 and the first-set restrictions support this, but unobservable within-set form could remain. Second, the data contain no direct measures of psychological states, so the mechanism discussion is interpretive and the three candidates are not mutually exclusive. Third, the design cannot separate the effect of winning on the winner from the effect of losing on the loser. Fourth, the analysis covers a single sport. The task-type framework yields a testable prediction: women should exhibit stronger momentum in other technique-intensive sports such as archery, shooting, and snooker, but not in sports where physical strength is the primary determinant of outcomes.

5 Conclusion

This paper studies psychological momentum and gender in elite table tennis, exploiting deuce sets as a natural experiment across 12,227 professional matches from 2014 to 2026. Winning a deuce set raises the probability of winning the next set by 8.8 percentage points overall ($p < 0.001$). The effect is more than twice as large for women (12.9 pp, a 27% gain on the female baseline) as for men (5.7 pp, an 11% gain), and the gender gap widens with deuce duration. Point-level streak evidence shows that the female advantage is absent at the shortest winning streaks and builds up as competitive pressure accumulates, matching a cognitive-affective rather than a mechanical or purely physiological channel.

These findings reverse the gender pattern documented in the prior literature on tennis and judo, where momentum benefits men but not women. A task-type framework reconciles the apparently conflicting evidence. In sports where physical endurance and strength matter, a physiological pathway from winning to subsequent performance favours men. In a technique-intensive sport like

table tennis, this pathway is muted and a cognitive-affective pathway dominates. Because women's competitive responses are more sensitive to context and feedback than men's, the cognitive-affective pathway then favours women. The framework generates a testable prediction: the female momentum advantage should extend to other technique-intensive sports such as archery, shooting, and snooker, but not to sports in which physical strength is the primary determinant of outcomes.

More broadly, the results show that gender differences in competitive dynamics are domain-specific, paralleling the evidence on gender differences in willingness to compete ([Wieland and Sarin, 2012](#); [Dreber et al., 2011](#); [Gneezy et al., 2009](#)) and in responses to high-stakes pressure ([Cai et al., 2019](#); [Iriberry and Rey-Biel, 2019](#); [De Paola and Scoppa, 2017](#); [Lackner and Weichselbaumer, 2023](#)). The implication extends beyond sport. If the psychological dynamics that convert winning into subsequent success depend on whether the task rewards physical aggression or fine-grained execution, uniform interventions to close competitive gender gaps will succeed in some domains and fail in others. Feedback-rich environments that reward precision rather than power, such as high-stakes examinations, skilled technical tasks, and tournament structures with repeated evaluation, may generate gender dynamics closer to the table tennis pattern than to the judo or tennis one. Mapping the set of tasks along this dimension is a first-order direction for research on gender and competition.

Declarations

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Declaration of Competing Interests

The author has no competing interests to declare.

Data Availability

The set-level panel used in this study was constructed from publicly available point-by-point match results published by Sofascore (<https://www.sofascore.com>). The constructed data and the Stata code that generates all tables and figures are available from the author upon reasonable request.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude in order to support language polishing. After using this tool, the author reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A Supplementary Results

A.1 Randomisation Verification

Table A1 verifies the identifying assumption behind the empirical strategy. The reference-player win rate in deuce sets is 49.5% overall and is indistinguishable from 50% across rounds, years, and first-set subsamples. The balance regression shows that *SetsAhead* is itself predictive of deuce-set outcomes, because players already leading the match close deuce sets at a slightly higher rate, so the main specifications condition on it. Once *SetsAhead* is held fixed, neither *Female* nor *China-involved* predicts the deuce outcome at conventional levels, and in the first-set subsample (where *SetsAhead* is mechanically zero) *Female* has a point estimate of 0.004 (SE = 0.028). Conditional quasi-randomness therefore holds in the regression that identifies the gender gap.

Table A1: Predicting Deuce Outcomes

	(1) All deuce	(2) + Controls	(3) + Set FE	(4) Set 1 only
<i>Dep. var.: Won deuce set</i>				
Sets ahead	0.046*** (0.005)	0.046*** (0.005)	0.043*** (0.006)	
Female		0.009 (0.012)	0.019 (0.014)	0.004 (0.028)
China involved		0.021 (0.014)	0.009 (0.016)	0.057* (0.033)
Round (1–8)			0.007 (0.005)	0.007 (0.011)
Set number FE	No	No	Yes	—
R-squared	0.010	0.011	0.011	0.004
Observations	7,501	7,501	5,512	1,269

Notes. Sample: sets that reached deuce (10–10+). Dependent variable: indicator for winning the deuce set. *Sets ahead* is mechanically nonzero for sets 2+. Robust standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

A.2 Momentum Gradient Across Margins

To contextualise the magnitude of deuce momentum, Figure A1 plots the set winner's next-set win probability across all margin categories: deuce (margin always 2), non-deuce margin 2 (e.g., 11–9), margin 3, margin 4, and margin 5+. The pattern is monotonically increasing: from 53.0% at deuce to 65.6% at margin 5+. This gradient shows that momentum scales with the decisiveness of the victory, and confirms that deuce sets, the most competitive possible set outcome, exhibit the smallest winner advantage, exactly as expected if quasi-randomness attenuates selection bias at narrow margins.

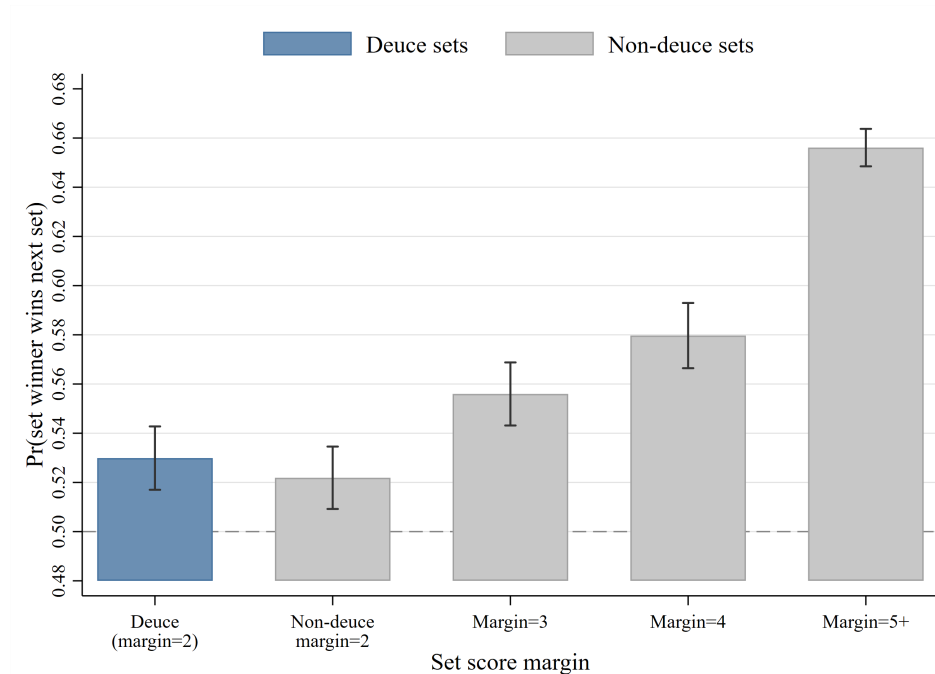


Figure A1: Winner Advantage by Set Margin

Notes. Each bar reports the set winner's probability of winning the next set. Teal: deuce sets (margin always 2). Gray: non-deuce sets by point margin. Error bars show 95% confidence intervals. The dashed line marks 0.50. Sample: sets with a subsequent set.

A.3 Player Fixed Effects

Table A2 re-estimates the main specifications using a player-level panel with player fixed effects. Each deuce set generates two observations (one per player), and all models absorb time-invariant

player ability. The momentum effect is 3.6 pp without controls (Column 1) and 6.1 pp with controls for sets ahead (Column 2). Columns (3) and (4) confirm the gender pattern: 9.7 pp for women ($p < 0.001$) and 3.4 pp for men ($p = 0.010$).

Table A2: Player Fixed Effects

	(1) Base	(2) Controls	(3) Female	(4) Male
<i>Dep. var.: player won next set</i>				
Won deuce set	0.036*** (0.010)	0.061*** (0.010)	0.097*** (0.014)	0.034** (0.013)
Sets ahead		0.059*** (0.005)	0.062*** (0.007)	0.056*** (0.007)
Player FE	Yes	Yes	Yes	Yes
Cluster SE	Player	Player	Player	Player
DV mean	0.505	0.505	0.503	0.506
Observations	11,235	11,235	4,804	6,431
Player clusters	885	885	389	496
Within R-squared	0.001	0.013	0.020	0.010

Notes. Unit of observation: player $\times$ deuce set. Dependent variable: indicator for winning the next set. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Robustness

Table A3 reports robustness checks. The baseline effect of 8.8 pp is stable across alternative specifications: first sets only (8.9), heteroskedasticity-robust standard errors without clustering (8.8, $p < 0.001$), best-of-five matches only (7.4), and excluding matches involving Chinese players (8.0).

Table A3: Robustness

	(1) Baseline	(2) Set 1	(3) HC1 SE	(4) Best-of-5	(5) No China
<i>Dep. var.: Won next set</i>					
Won deuce set	0.088*** (0.013)	0.089*** (0.024)	0.088*** (0.013)	0.074*** (0.016)	0.080*** (0.015)
Sets ahead	0.072*** (0.007)		0.072*** (0.007)	0.071*** (0.009)	0.064*** (0.008)
Female	-0.019 (0.013)	-0.053** (0.024)	-0.019 (0.013)	-0.045*** (0.016)	-0.026* (0.015)
China involved	-0.012 (0.016)	0.016 (0.029)	-0.012 (0.016)	-0.039* (0.020)	
Cluster SE	Match	Match	HC1	Match	Match
DV mean	0.495	0.493	0.495	0.484	0.497
Observations	5,773	1,784	5,773	4,145	4,544
Match clusters	4713	1784	.	3481	3716
R-squared	0.023	0.011	0.023	0.019	0.018

Notes. Sample: deuce sets with a subsequent set. Dependent variable: indicator for winning the next set. Standard errors in parentheses, clustered by match unless noted. * p<0.10, ** p<0.05, *** p<0.01.

A.5 Match-Level Outcome

Table A4 replaces the next-set outcome with the match outcome, testing whether deuce momentum carries through to the final result. Winning a deuce set increases the probability of winning the match by a large and significant margin when restricting to non-final sets, confirming that momentum extends well beyond the immediate next set. The gender interaction remains significant: women gain an additional 4.8 pp relative to men ($p = 0.029$). Figure A2 visualises the gender gap in match-level carry-over by deuce duration, complementing the set-level dose–response of Figure 1.

Table A4: Match-Level Outcome

	(1) All	(2) Non-final sets only	(3) Interaction (all)	(4) Interaction (non-final)	(5) Triple (non-final)
<i>Dep. var.: Won match</i>					
Won deuce set	0.398*** (0.009)	0.355*** (0.011)	0.383*** (0.011)	0.334*** (0.014)	0.339*** (0.026)
Won $\times$ Female			0.036** (0.017)	0.048** (0.022)	0.026 (0.039)
Won $\times$ Duration					-0.001 (0.006)
Won $\times$ Female $\times$ Duration					0.005 (0.008)
Final sets included	Yes	No	Yes	No	No
Lower-order terms	—	—	Yes	Yes	Yes
Sets ahead control	Yes	Yes	Yes	Yes	Yes
DV mean	0.498	0.496	0.498	0.496	0.496
Observations	7,501	5,773	7,501	5,773	5,773
Match clusters	5728	4713	5728	4713	4713
R-squared	0.371	0.236	0.372	0.236	0.237

Notes. Sample: deuce sets. Dependent variable: indicator for winning the match. Final-set deuces are mechanically won by the deuce winner and are excluded where the *Final sets included* row is marked . Standard errors in parentheses, clustered by match. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

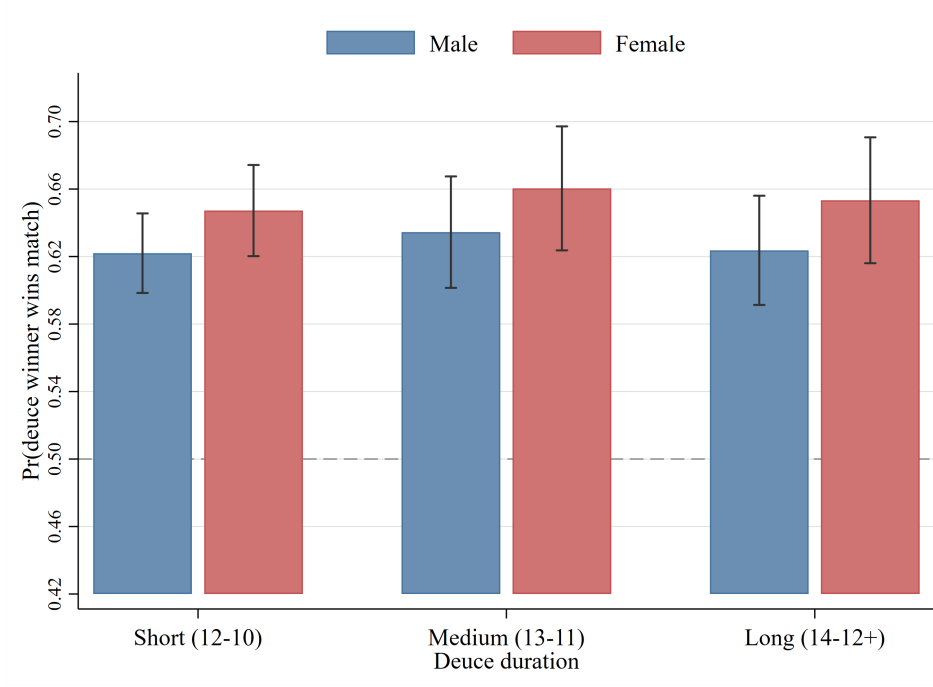


Figure A2: Match-Level Momentum by Gender and Deuce Duration

Notes. Each bar reports the deuce-set winner's probability of winning the match, by deuce duration and gender. Error bars show 95% confidence intervals. The dashed line marks 0.50. Sample: non-final deuce sets (final-set deuces are mechanically won by the deuce winner).

A.6 Micro-Momentum at $k = 1$: Ruling Out Mechanical Artefacts

Table A5 supports the point-level discussion in Section 3.5 by re-estimating the probability of winning the next point in deuce sets, restricted to points following a single-point winning streak ($k = 1$). If the set-level gender gap were driven by mechanical features of the game (serve rotation, scoring rules, or simple mean reversion), it should appear at $k = 1$ with similar magnitude. The estimated gender gap at $k = 1$ is small and statistically indistinguishable from zero, ruling out this class of explanations and leaving room for a channel whose effect accumulates with streak length.

Table A5: Micro-Momentum at $k = 1$: Ruling Out Mechanical Artifacts

	(1) Pooled	(2) Male	(3) Female	(4) Interaction
<i>Dep. var.: Won current point</i>				
Prev. point won	-0.028*** (0.009)	-0.029** (0.012)	-0.027** (0.013)	-0.029** (0.012)
Female				-0.009 (0.012)
Prev. won $\times$ Female				0.001 (0.018)
Constant	0.505*** (0.006)	0.509*** (0.008)	0.500*** (0.009)	0.509*** (0.008)
Observations	13,819	7,766	6,053	13,819
R-squared	0.001	0.001	0.001	0.001

Notes. Sample: points in the deuce phase with a lagged outcome available. Dependent variable: indicator for the home player winning the current point. *Prev. won* is an indicator for the home player having won the immediately preceding point (streak length $k = 1$). Standard errors in parentheses, clustered by match. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.7 Closing Streaks and the Female Advantage

Table A6 links the within-set streak structure to the set-level outcome. I partition deuce sets by the length k of the closing streak, defined as the number of consecutive final points won by the eventual set winner immediately before the set ended, and within each partition I estimate the female–male gap in next-set win probability. The gap is positive in the “battle” bin where the deuce is closed on a one-point streak ($k = 1$, +9.5 pp, $p < 0.05$, $N = 567$), indistinguishable from zero in the middle bins ($k = 2$ and $k = 3$, $N = 1,043$ and 772), and positive again in the “dominant” bin where the closing streak runs four or more points ($k \geq 4$, +11.3 pp, $p < 0.10$, $N = 241$). The pattern is not monotone, and the small sample in the $k \geq 4$ cell makes that bin exploratory. The exercise is therefore a robustness check rather than a mechanism test: it shows that the female advantage survives conditioning on the within-set closing structure rather than being an artefact of the aggregate set-level analysis.

Table A6: Bridge: Closing Streak and Gender Gap in Next-Set Win

	(1) $k = 1$ (battle)	(2) $k = 2$	(3) $k = 3$	(4) $k \geq 4$ (dominant)
<i>Dep. var.: Winner wins next set</i>				
Female	0.095** (0.044)	-0.012 (0.030)	0.036 (0.036)	0.113* (0.063)
Constant (Male)	0.464*** (0.026)	0.554*** (0.020)	0.508*** (0.024)	0.544*** (0.043)
Observations	567	1,043	772	241
R-squared	0.009	0.000	0.001	0.013

Notes. Sample: deuce sets with a subsequent set. Dependent variable: indicator for the deuce-set winner winning the next set. Each column restricts to deuce sets closed on a winning streak of length k (the final consecutive points won by the eventual set winner). Standard errors in parentheses, clustered by match. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.